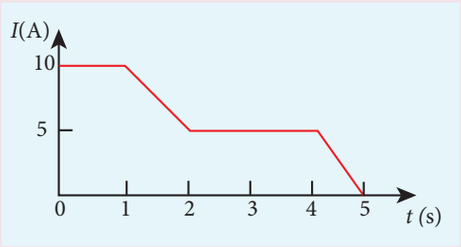
7. The current through an element is shown in the figure. Determine the total charge that pass through the element at a) t = 0 s, b) t = 2 s, c) t = 5s



Ans: At t= 0s,dq = 0 C, At t=2 s, dq = 10 C; At t=5 s, dq = 0 C

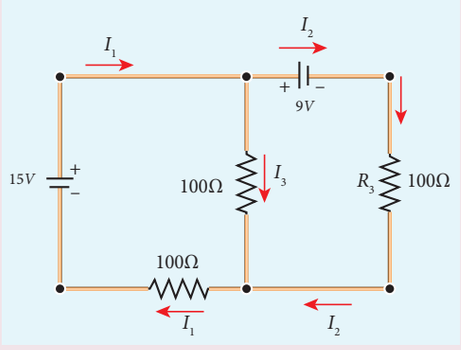
8. An electronics hobbyist is building a radio which requires 150 Ω in her circuit, but she has only 220 Ω, 79 Ω and 92 Ω resistors available. How can she connect the available resistors to get desired value of resistance?

Ans: Parallel combination of 220 Ω and 79 Ω in series with 92 Ω

9. A cell supplies a current of 0.9 A through a 2 Ω resistor and a current of 0.3 A through a 7 Ω resistor. Calculate the internal resistance of the cell.

Ans: 0.5 Ω

10. Calculate the currents in the following circuit.

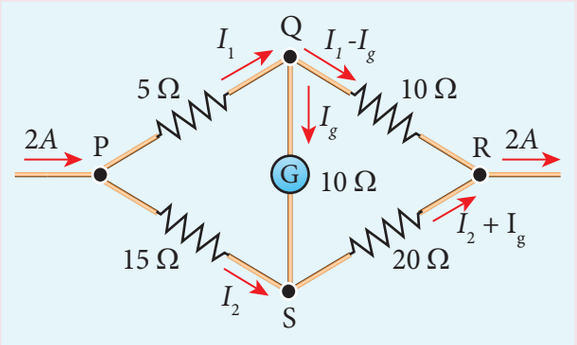


Ans : I_1 = 0.070 A, I_2 = -0.010 A and I_3 = 0.080 A

11. A potentiometer wire has a length of 4 m and resistance of 20 Ω. It is connected in series with resistance of 2980 Ω and a cell of emf 4 V. Calculate the potential along the wire.

Ans: Potential = 0.65×10^{-2} V m^{-1}.

12. Determine the current flowing through the galvanometer (G) as shown in the figure.



Ans: $I_g = \frac{1}{11} A$

13. Two cells each of 5V are connected in series across a 8 Ω resistor and three parallel resistors of 4 Ω, 6 Ω and 12 Ω. Draw a circuit diagram for the above arrangement. Calculate i) the current drawn from the cell (ii) current through each resistor

Ans: The current at 4 Ω , $I = \frac{2}{4} = 0.5A$,

the current at 6 Ω, $I = \frac{2}{6} = 0.33\,\text{A}$,

the current at 12 Ω, $I = \frac{2}{12} = 0.17\,\text{A}$

14. Four light bulbs P, Q, R, S are connected in a circuit of unknown arrangement. When each bulb is removed one at a time and replaced, the following behavior is observed.

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